Error handling in type script using Try-catch

Error handling is a mechanism used to detect and handle errors that occur while a program is running. Instead of stopping the entire program when an error occurs, we can handle the error and display a suitable message.
TypeScript uses the JavaScript try-catch mechanism for error handling.

Syntax

```
try {
   // Code that may cause an error
} catch (error) {
   // Code to handle the error
}
```

How it works

1. **try block** – Contains code that may generate an error.
2. **catch block** – Executes when an error occurs in the try block.
3. **error** – Contains information about the error.

Example 1: Basic try-catch

```
let num1: number = 10;
let num2: number = 0;

try {
   if (num2 === 0) {
      throw new Error("Cannot divide by zero");
   }

   let result: number = num1 / num2;
   console.log("Result:", result);
}
catch (error) {
   console.log("Error:", (error as Error).message);
}
```

Output:

Error: Cannot divide by zero

Here, throw is used to create an error when num2 is zero. The catch block catches the error and displays its message.

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Example 2: Handling invalid array index

```
let students: string[] = ["Rahul", "Amit", "Neha"];

try {
   let index: number = 5;

   if (index >= students.length) {
      throw new Error("Student index is out of range");
   }

   console.log(students[index]);
}
catch (error) {
   console.log("Error:", (error as Error).message);
}
```

Output:

Error: Student index is out of range

Using finally

TypeScript also supports the **finally** block. The finally block executes whether an error occurs or not.

```
try {
   console.log("Program is running...");
   throw new Error("Something went wrong");
}
catch (error) {
   console.log("Error:", (error as Error).message);
}
finally {
   console.log("Program execution completed.");
}
```

Output:

Program is running...
Error: Something went wrong
Program execution completed.

Keyword	Purpose
try	Contains code that may cause an error
catch	Handles the error
throw	Creates/raises an error
finally	Executes regardless of whether an error occurs

In short: try-catch makes TypeScript programs more reliable by allowing errors to be detected and handled without abruptly terminating the program.